

OOP MCQ CHALLENGE MODE (LEVEL SYSTEM)

LEVEL 1: EASY WARM-UP (No silly mistakes allowed)

Q1. Which is NOT an OOP feature?

- a) Encapsulation
- b) Inheritance
- c) Polymorphism
- d) Compilation

 **Your brain says?**

 *Think: OOP is about design, not process...*

 **Answer:** d) Compilation

Q2. Encapsulation means:

- a) Wrapping data + functions
- b) Creating classes
- c) Hiding only data
- d) Inheritance

 *Hint: Capsule = everything packed together*

 **Answer:** a)

Q3. Default access specifier in class?

- a) Public
- b) Private
- c) Protected
- d) None

 *Trap question (very common in exams)*

 **Answer:** b) Private

LEVEL 2: CONCEPT FIGHT (Now it gets real)

Q4. What is polymorphism?

- a) Same function → different behavior

- b) Data hiding
- c) Code reuse
- d) Object creation

💡 Think: *Poly = many forms*

✅ **Answer:** a)

Q5. Which is NOT valid in C++?

- a) Public
- b) Private
- c) Protected
- d) Friendly

😂 C++ is not that friendly bro

✅ **Answer:** d)

Q6. Constructor purpose?

- a) Destroy object
- b) Initialize object
- c) Create class
- d) None

🔥 Easy but important

✅ **Answer:** b)

💀 **LEVEL 3: CODE OUTPUT (Brain damage zone)**

Q7. What will this print?

```
Base* b;  
Derived d;  
b = &d;  
b->show();
```

(No virtual used)

- a) Base
- b) Derived
- c) Error
- d) Runtime error

🧠 Rule: *No virtual* → *early binding*

✅ **Answer:** a) Base

Q8. What will this print?

```
Base* b = new Derived();  
b->print(); // virtual
```

- a) Base
- b) Derived
- c) Error
- d) Crash

🔥 Rule: *Virtual* = *runtime magic*

✅ **Answer:** b) Derived

🧠 LEVEL 4: MASTER CONCEPTS

Q9. Pure virtual function means:

- a) No implementation
- b) Must override
- c) = 0 syntax
- d) All

💀 If you miss this → danger

✅ **Answer:** d)

Q10. Abstract class means:

- a) Cannot create object
- b) Has pure virtual

- c) Both
- d) None

✓ **Answer: c)**

Q11. Constructor order?

- a) Derived → Base
- b) Base → Derived
- c) Random
- d) None

🔥 Golden rule

✓ **Answer: b)**

⚡ LEVEL 5: TRICKY + INTERVIEW LEVEL

Q12. Virtual destructor is used for:

- a) Speed
- b) Cleanup properly
- c) Overloading
- d) Nothing

🧠 Think: deleting derived via base pointer

✓ **Answer: b)**

Q13. this pointer refers to:

- a) Global object
- b) Current object
- c) Static variable
- d) None

✓ **Answer: b)**

Q14. Operator overloading allows:

- a) New operators

- b) Use operators on objects
- c) Change syntax
- d) None

🔥 Real-world concept

✅ **Answer: b)**

🏆 FINAL BOSS ROUND (ULTIMATE RULES)

🎯 If you remember ONLY this → you pass exam:

- 🔥 **No virtual → Base class output**
 - 🔥 **Virtual → Derived class output**
 - 🔥 **Constructor → Base first**
 - 🔥 **Destructor → Derived first**
 - 🔥 **Encapsulation = Binding + Hiding**
 - 🔥 **Polymorphism = Same name, different work**
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🎯 BONUS CHALLENGE (TRY YOURSELF)

👉 What will this print?

```
class A {  
public:  
    virtual void show() { cout << "A"; }  
};
```

```
class B : public A {  
public:  
    void show() { cout << "B"; }  
};
```

```
A* a = new B();  
a->show();
```

 Think before scrolling...



 **Answer: B**